Climate Response of Second-Season Corn in Brazil

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Abstract

Brazil's second-season corn is the country's dominant corn season and a central component of global grain supply, but its short post-soybean window concentrates exposure to heat and water stress. We estimate yield responses across Brazil's major producing regions from 2003 to 2023 using municipal yield data, daily gridded weather, and region-specific phenological calendars. In the main specification, one additional killing degree day above 29°C lowers yield by about 0.10%. Heat damage is concentrated in early grain-fill exposure, while water-deficit effects are most pronounced during flowering and early grain fill. Regional estimates show the largest heat penalties in Goiás, Minas Gerais, Paraná, and Mato Grosso do Sul, while Mato Grosso's conditional response is statistically indistinguishable from zero. A uniform warming counterfactual reduces predicted yield levels by 8.2% under 1°C, 16.4% under 2°C, and 24.4% under 3°C.

Highlights

- Extreme heat lowers second-season corn yields by about 0.10% per KDD.
- Heat damage peaks in early grain fill; water-deficit damage in flowering and early grain fill.
- A uniform 1°C warming counterfactual reduces predicted yield by 8.2%.

Keywords: second-season corn; climate change; extreme heat; water deficit; crop phenology; Brazil

JEL classifications: Q12; Q54; O13

1 Introduction

Brazil is the world's second-largest corn exporter (USDA Foreign Agricultural Service, 2026). Around 80% of its corn is grown as a second crop (CONAB, 2025). In the major central producing regions, producers typically harvest soybeans in January–February and plant corn immediately into the same field, with second-season corn harvested mainly between June and August (CONAB, 2025; Pires et al., 2016). This second crop, known in Brazil as safrinha, is now the dominant corn season, having overtaken traditional first-season planting after the early 2000s (see Figure 1).

Safrinha is planted into the dry-season tail of central Brazil. By the time the crop reaches its critical reproductive stages, regional rainfall is declining and evapotranspiration remains high. Safrinha therefore reaches flowering and grain fill as rainfall recedes, making water availability during those stages a central production constraint (Andrea et al., 2019; Magalhães et al., 2002). This post-soybean system loads weather risk onto the second crop in a way the first crop does not face. Because it is planted after soybean, safrinha advances farther into the retreating rainy season and leaves less timing margin than the first crop (Pires et al., 2016). This difference in seasonal timing could potentially help explain the higher yield volatility of second-season corn relative to first-season corn shown in Figure 1.

The agronomic literature on corn is clear that the impact of weather on yields depends on phenological stage. Heat damage is often strongest around flowering, when even brief exposure to extreme temperatures can reduce pollination, kernel set, and final yield; heat during early grain fill can also reduce yield, and the response differs materially from vegetative and late-stage exposure (Cicchino et al., 2010; Li et al., 2022; Liu et al., 2023; Dong et al., 2021). County-panel evidence for U.S. maize likewise finds that average extreme-

temperature sensitivity peaks during silking and grain filling, with KDD sensitivity generally largest during early grain fill (Butler and Huybers, 2015). Water stress follows a similar logic because water availability during flowering and grain fill is a central determinant of yield, and those stages govern pollination, kernel set, and grain accumulation (Magalhães et al., 2002; Andrea et al., 2019; Butts-Wilmsmeyer et al., 2019; Prudente Junior et al., 2024). Average-temperature and total-precipitation measures over the full season pool exposure across these biologically distinct phases, which can mask the nonlinearity and timing that drive yield outcomes (Ortiz-Bobea, 2013; Ortiz-Bobea et al., 2019; Wen et al., 2026).

Despite its economic importance, literature on safrinha is not vast. Cohn et al. (2016) estimate climate responses for double cropping in Mato Grosso (MT) over 2002/03–2009/10 and find that changes in cropping frequency and cropland area together account for roughly 70% of climate-related output changes and exceed yield responses. Their work documents important behavioral margins, but limits the climate analysis to a single state. Mato Grosso remains the largest safrinha producer, accounting for roughly 45% of national production in 2023 (IBGE, 2024), but other producing regions together account for more than half of the system. Ignoring those regions can yield estimates that are not representative of Brazil's current second-season corn geography, which spans Paraná (PR), Goiás (GO), Mato Grosso do Sul (MS), Minas Gerais (MG), São Paulo, and MATOPIBA.[1] These regions span a wide climatic gradient. Paraná lies in the subtropical South, while MT and MATOPIBA are lower-latitude regions where double cropping is more tightly constrained by rainy-season length and timing (Andrea et al., 2019; Abrahão and Costa, 2018).

Agronomic studies of safrinha based on process-based crop models project substantial yield losses under future warming (Andrea et al., 2019; Hampf et al., 2020). Recent national-scale forecasting work for Brazilian off-season maize likewise emphasizes rainfall, temperature, soil water, and reproductive-stage indicators as the variables most closely linked to interannual yield variation (Prudente Junior et al., 2024). A systematic review of this

[1] MATOPIBA comprises selected municipalities in Maranhão, Tocantins, Piauí, and Bahia rather than the entirety of those states (IBGE, 2024, 2022).

literature finds corn yields declining across climate scenarios, with the subtropical south less affected than the northern regions (Bigolin and Talamini, 2024). None of this work estimates causal climate sensitivity using observational variation in weather.

This paper estimates how second-season corn yields respond to moderate heat, extreme heat, and water deficit across phenological windows and producing regions in Brazil. The analysis has three central objectives. First, we quantify the sensitivity of safrinha yields to extreme heat and water deficit, allowing for biological heterogeneity across the phenological cycle. Second, we ask whether this sensitivity is heterogeneous across producing regions, given the wide climatic gradient spanned by the Brazilian safrinha system. Third, we translate the estimated response functions into 1°C, 2°C, and 3°C uniform warming counterfactuals to quantify how warmer daily temperature distributions affect second-season corn yields.

To address these objectives, we construct a municipality-year panel covering 1,518 municipalities over 2003–2023 (17,713 observations) in the seven main producing regions. We match daily gridded weather (Xavier et al., 2022) to municipality-specific safrinha phenology drawn from *Companhia Nacional de Abastecimento* (CONAB) regional planting calendars and decompose weather effects into five phenological windows, consisting of pre-planting (preceding the second-season corn cycle), vegetative, flowering, early grain-fill, and late grain-fill. The window definitions follow the agronomic and crop-modeling literature that evaluates maize weather exposure around emergence, flowering, grain fill, and maturity when individual field phenology is not observed at large geographic scales (Butts-Wilmsmeyer et al., 2019; Prudente Junior et al., 2024). Heat exposure is measured by growing and killing degree days (Butler and Huybers, 2013), while water deficit is measured by the difference between potential evapotranspiration and precipitation. The specification absorbs municipality and year fixed effects, together with region-specific linear time trends to capture region-specific technical change.

Our results indicate that extreme heat and water stress substantially reduce safrinha

yields. One additional extreme degree day lowers yield by about 0.10%, with the clearest damage during vegetative growth and early grain fill. Water-deficit losses are most evident during flowering and early grain fill. The response is also regionally heterogeneous. Goiás, Minas Gerais, Paraná, and Mato Grosso do Sul carry the largest heat penalties, while Mato Grosso's conditional extreme-heat response is small and statistically indistinguishable from zero. In a uniform warming counterfactual, a 1°C shift reduces predicted yield levels by 8.2%, with larger losses under 2°C and 3°C warming.

The paper proceeds as follows. Section 2 describes the development and current geography of safrinha in Brazil. Section 3 presents the materials and methods. Section 4 describes the panel construction and weather variables. Section 5 presents the national and regional response estimates. Section 6 translates the estimated response into uniform warming counterfactuals. Section 7 presents robustness checks. Section 8 discusses interpretation, limitations, and implications, and Section 9 concludes.

2 Brazil's Second-Season Corn System

Brazilian agriculture transformed in the 1990s and 2000s through the combination of soybean expansion in the Cerrado and the development of short-cycle maize hybrids suitable for sowing immediately after soy harvest (Pires et al., 2016; Jeddi and DePaula, 2026). Before the early 2000s, second-season corn accounted for only a small share of Brazil's corn area and production (Moreira Guimarães et al., 2022; Jeddi and DePaula, 2026). As Roundup-Ready soybean varieties shortened soy cycles and short-cycle corn hybrids became available, producers began planting a second crop into the residual moisture and warmth following the soybean harvest. By the late 2010s, safrinha had overtaken the traditional first-season planting in national production, and today it accounts for roughly 80% of Brazil's corn output (Moreira Guimarães et al., 2022; Jeddi and DePaula, 2026; CONAB, 2025).

Spatially, the early second-season crop was small and dispersed, but expansion after the soybean boom was concentrated in the Center-West (Moreira Guimarães et al., 2022; Jeddi

and DePaula, 2026). Mato Grosso emerged as the dominant producer in the 2000s. In 2003, PR, MS, and MT together accounted for 82.4% of national second-season corn production. By 2023, MT alone supplied around 45%, with Goiás, the MATOPIBA frontier, and Minas Gerais each contributing meaningful shares (IBGE, 2024). MATOPIBA is a newer agricultural frontier formed by an official set of municipalities within Maranhão, Tocantins, Piauí, and Bahia, not by those four states in their entirety (Moreira Guimarães et al., 2022; IBGE, 2022). Its suitability for double cropping is especially sensitive to wet-season length and timing (Abrahão and Costa, 2018).

The agronomic challenge of the safrinha window is that the critical reproductive stages of maize coincide with the end of the wet season and the early dry season in central Brazil. Producers must time planting to ensure that flowering and grain-fill occur before the dry season severely limits soil moisture and increases atmospheric demand. In practice, this means planting as early as possible after the soy harvest, ideally before the end of February in MT and Goiás (CONAB, 2025; Pires et al., 2016). Producers planting later face higher risk of water deficit and heat exposure during grain-fill. Andrea et al. (2019) document substantial yield penalties for planting outside the recommended window in MT.

Figure 1 summarizes the scale of this transformation. Between 2003 and 2023, safrinha harvested area increased from 3.3 to 17.3 million ha, while safra area fell from 9.7 to 5.0 million ha. Production shifted even more sharply, with safrinha production rising from 13.3 to 104.3 million tons while safra production declined from 35.0 to 27.6 million tons. Yield levels are much closer than the area and production series, but safrinha yields fluctuate more from year to year, with especially sharp downturns in 2004–2005, 2016, and 2021. This contrast is consistent with a system that has expanded rapidly into the post-soybean window while remaining exposed to interannual weather risk.

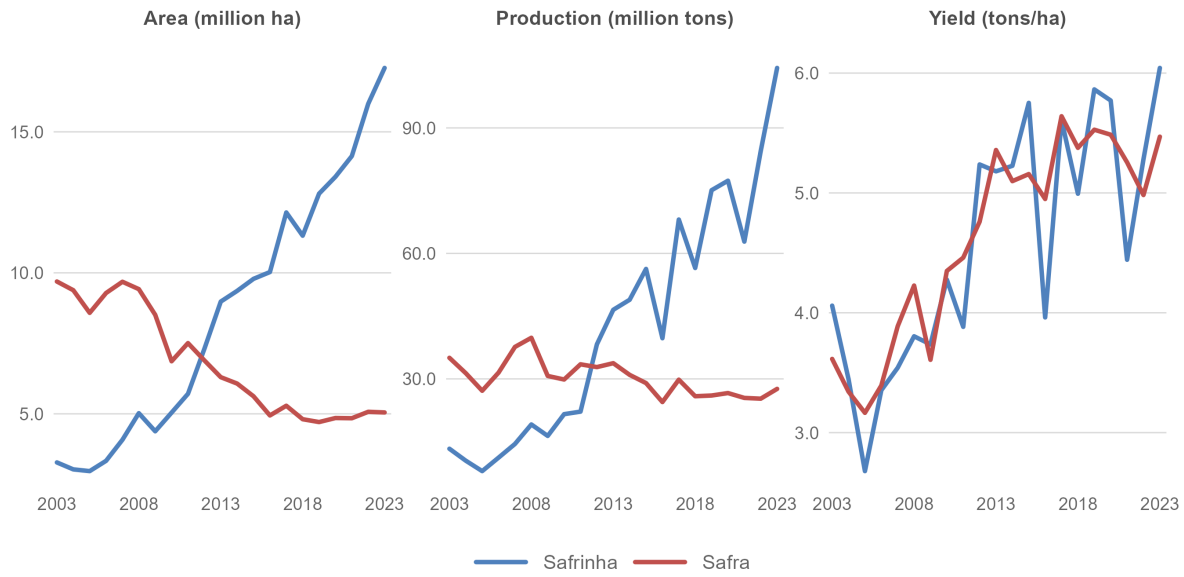


Figure 1: Evolution of first-season corn (safra) and second-season corn (safrinha) area, production, and yield in Brazil, 2003–2023. Source: authors' calculations using IBGE PAM (IBGE, 2024).

3 Materials and Methods

3.1 Regression Specification

We estimate the climate response of log yield using a fixed-effects panel regression with phenologically-resolved weather:

$$\begin{aligned} \log Y_{mt} = \alpha_m + \lambda_t + \rho_{r(m)} \cdot t \\ + \sum_{w \in \mathcal{W}^{\text{cyc}}} \left(\beta_w^{\text{GDD}} \text{GDD}_{wmt} + \beta_w^{\text{KDD}} \text{KDD}_{wmt} \right) \\ + \sum_{w \in \mathcal{W}} \left(\gamma_w \text{WD}_{wmt} + \delta_w \text{WD}_{wmt}^2 \right) + \varepsilon_{mt}. \end{aligned} \tag{1}$$

where m indexes municipality, t indexes agricultural year, and $r(m)$ is the region containing m. The phenological windows are denoted by $\mathcal{W}$, and include pre-planting, vegetative, flowering, early grain-fill, and late grain-fill. On the other hand, $\mathcal{W}^{\text{cyc}}$ includes all windows except pre-planting.

Heat enters the regression as a linear function of Growing Degree Days (GDD) and Killing Degree Days (KDD) per yield-cycle window. We define moderate heat as GDD between 8

and 29°C, and extreme heat as KDD above 29°C (Schlenker and Roberts, 2009; Butler and Huybers, 2013). Pre-planting heat is excluded because the crop is not yet in the field. Water deficit (WD) enters as a quadratic per window for all five windows, including pre-planting. The pre-planting WD terms are included as a control because preceding moisture conditions may affect planting and stand establishment through the water available at germination, but are not part of the yield-cycle moisture response we wish to estimate (Cohn et al., 2016; Magalhães et al., 2002).

The model also includes municipality fixed effects (α_m), year fixed effects (λ_t), and region-specific linear time trends ($\rho_{r(m)} \cdot t$), with t centered at 2013. Regions are MATOPIBA, defined by the official list of 337 municipalities in Maranhão, Tocantins, Piauí, and Bahia (IBGE, 2022), and the state-based regions Mato Grosso (MT), Goiás (GO), Minas Gerais (MG), Mato Grosso do Sul (MS), São Paulo (SP), and Paraná (PR). Municipalities outside these seven producing regions are excluded from the estimation sample. Standard errors are clustered at the state-year level.

Identification relies on the fact that, conditional on municipality fixed effects, weather variables enter the regression as deviations from each municipality's long-run mean. That is, they are transformed into year-to-year weather shocks, which are plausibly random and exogenous (Dell et al., 2014; Blanc and Schlenker, 2017).

In addition, the fixed-effects structure addresses several confounders. Municipality fixed effects absorb time-invariant soil quality, baseline yield, and farmer characteristics. Year fixed effects absorb shocks common to all municipalities in a given year, including national policy, prices, and country-wide weather anomalies. Finally, region-specific time trends absorb the differential pace of varietal improvement, mechanization, and input adoption across regions. This is important because crop composition, farm structure, technology, and the recent agricultural transformation differ substantially across Brazilian regions (Castro et al., 2020; Jeddi and DePaula, 2026).

3.2 Derived Quantities for Display

The regression in (1) delivers eight heat coefficients (one per window per channel) and ten water-deficit coefficients (linear and quadratic per window). To summarize the heat response into a single number comparable across regions and specifications, we report the weighted-average marginal heat damage,

$$\bar{\beta}^{\text{KDD}} = \frac{\sum_{w \in \mathcal{W}^{\text{cyc}}} \bar{m}_w \cdot \beta_w^{\text{KDD}}}{\sum_{w \in \mathcal{W}^{\text{cyc}}} \bar{m}_w}, \tag{2}$$

where $\bar{m}_w$ is the sample-mean KDD exposure in window w. This is the linear combination of per-window coefficients that would obtain if a marginal extreme-heat day were distributed across windows in proportion to their typical exposure. We compute the analogous quantity within each region by replacing the pooled per-window coefficients with their region-specific counterparts.

For water deficit, the marginal effect varies with the level of WD because the response is quadratic. To convert the per-window WD coefficients into a single marginal-effect curve as a function of total cycle WD, we follow a two-step procedure. First, for each window w in $\mathcal{W}^{\text{cyc}}$ we fit an auxiliary OLS regression of within-window WD on the total cycle WD,

$$\text{WD}_{wmt} = \alpha_w^a + \phi_w \cdot \text{WD}_{mt}^{\text{cyc}} + \varepsilon_{wmt}^a, \tag{3}$$

where $\text{WD}_{mt}^{\text{cyc}} = \sum_{w \in \mathcal{W}^{\text{cyc}}} \text{WD}_{wmt}$ is the total yield-cycle water deficit. By construction $\sum_w \phi_w = 1$ and $\sum_w \alpha_w^a = 0$. The auxiliary regression simply decomposes how a marginal mm of total cycle WD distributes across windows in the data. Second, conditional on a level x of total cycle WD, the predicted per-window WD is $\widehat{\text{WD}}_w(x) = \alpha_w^a + \phi_w x$. Substituting into the per-window damage formula gives the per-window-summed marginal effect of total cycle WD,

$$\text{Marg}_{\text{PW}}(x) = \sum_{w \in \mathcal{W}^{\text{cyc}}} \phi_w \left(\gamma_w + 2\delta_w \cdot \widehat{\text{WD}}_w(x) \right), \tag{4}$$

where γ_w and δ_w are the coefficients of WD_w and WD_w^2, respectively, from Equation (1). Substituting $\widehat{\mathrm{WD}}_w(x) = \alpha_w^a + \phi_w x$ and collecting terms in x, the curve is linear in x:

$$\mathrm{Marg}_{\mathrm{PW}}(x) = A + Bx, \tag{5}$$

with intercept and slope

$$A = \sum_{w \in \mathcal{W}^{\mathrm{cyc}}} \phi_w \gamma_w + 2 \sum_{w \in \mathcal{W}^{\mathrm{cyc}}} \phi_w \alpha_w^a \delta_w, \qquad B = 2 \sum_{w \in \mathcal{W}^{\mathrm{cyc}}} \phi_w^2 \delta_w. \tag{6}$$

Because $\phi_w \in [0, 1]$ and $\sum_w \phi_w = 1$, the squared weights ϕ_w^2 are bounded, so the slope B is well-behaved regardless of the sign of the window-level water deficit. This avoids the numerical instability that affects a naive proportional-allocation construction when window WDs span both signs. Standard errors for $\mathrm{Marg}_{\mathrm{PW}}(x)$ are computed by the delta method, treating the auxiliary regression coefficients (α_w^a, ϕ_w) as known constants. Regional versions replace the pooled per-window WD coefficients and the auxiliary coefficients with their region-specific counterparts.

For comparison with the existing literature, we also report a growing-season-pooled (GS) specification that replaces the per-window regressors with their sums across yield-cycle windows:

$$\begin{aligned} \log Y_{mt} = \alpha_m + \lambda_t + \rho_{r(m)} \cdot t \\ + \beta_{\mathrm{GS}}^{\mathrm{GDD}} \mathrm{GDD}_{mt}^{\mathrm{cyc}} + \beta_{\mathrm{GS}}^{\mathrm{KDD}} \mathrm{KDD}_{mt}^{\mathrm{cyc}} \\ + \gamma_{\mathrm{GS}} \mathrm{WD}_{mt}^{\mathrm{cyc}} + \delta_{\mathrm{GS}} (\mathrm{WD}_{mt}^{\mathrm{cyc}})^2 \\ + \tilde{\gamma} \mathrm{WD}_{mt}^{\mathrm{pre}} + \tilde{\delta} (\mathrm{WD}_{mt}^{\mathrm{pre}})^2 + \varepsilon_{mt}. \end{aligned} \tag{7}$$

The GS-pooled specification corresponds to the standard approach in the climate economics literature that aggregates exposure over a single growing-season window (Schlenker and Roberts, 2009). We use Equation (7) as a benchmark. The phenologically-resolved specification (Equation (1)) is the main specification.

4 Data

This section describes the four data sources used to construct the estimation panel: municipal yield and area, gridded daily weather, phenological calendars, and the heat and water deficit measures derived from them. We then describe sample construction.

4.1 Yield and Area

Annual municipal yield and area data come from the Brazilian Institute of Geography and Statistics (IBGE) *Pesquisa Agrícola Municipal* (PAM) (IBGE, 2024), which reports planted area, harvested area, and production for second-season maize (*milho 2ª safra*) at the municipality–year level. PAM has separately tracked first- and second-crop maize since 2003, defining the start of our panel. We compute yield as the ratio of production to harvested area in tonnes per hectare and use its natural logarithm as the outcome variable.

4.2 Weather

Daily gridded weather data come from the Brazilian Daily Weather Gridded Data (BR-DWGD) version 3.2.3 (Xavier et al., 2022; Xavier, 2024), a $0.1° \times 0.1°$ product covering January 1961 through March 20, 2024. We use daily minimum and maximum temperature, precipitation, and the variables required to compute reference evapotranspiration following Penman-Monteith (Allen et al., 1998). The grid is aggregated to municipality-day means using the IBGE 2020 municipality shapefile, with area-weighted averaging across grid cells overlapping each municipality.

Because the mapping from daily temperature to degree days is nonlinear, heat exposure measures are computed at the daily grid-cell level and then averaged to the municipality-day level, rather than first averaging temperature across the grid and then computing degree days from a smoothed daily mean. This preserves the within-municipality distribution of temperature extremes that drives the heat-exposure measures (Schlenker and Roberts, 2009; Butler and Huybers, 2013).

4.3 Phenological Windows

We define five phenological windows for each municipality-year based on the typical phenology of short-cycle maize grown in Brazil (CONAB, 2025; Magalhães et al., 2002; Li et al., 2022). Let day 0 denote the municipality-specific safrinha planting date. We obtain planting dates from the mesorregion-level phenological calendar published by the *Companhia Nacional de Abastecimento* (CONAB), which since the 2018/2019 season has issued bi-weekly crop-progress reports documenting the typical phenological stage of second-season corn in producing mesorregions. We match each municipality in the sample to its CONAB mesorregion; municipalities without a direct mesorregion match are assigned the calendar of the nearest covered mesorregion by geographic centroid.[2]

The five windows are:

- **Pre-planting** (-90 to 0): the three months preceding safrinha planting, corresponding to the soy growing season and the moisture-recharge period before safrinha sowing.
- **Vegetative** (0 to 46): planting through 14 days before silking.
- **Flowering** (46 to 74): silking ± 14 days; the critical pollination window.
- **Early grain-fill** (74 to 105): R3 (milk) through R5 (dent) developmental stages.
- **Late grain-fill** (105 to 130): R5 through R6 (physiological maturity).

We refer to the four post-planting windows collectively as the *yield-cycle windows*. Pre-planting water deficit is included in the regression as a control because preceding moisture conditions may affect planting and stand establishment through the water available at germination, but are not part of the yield-cycle moisture response we wish to estimate (Cohn et al., 2016; Magalhães et al., 2002). This window structure follows Ortiz-Bobea (2013) in aligning the regression's temporal aggregation with maize developmental physiology rather than fixed calendar months.

[2] Mesorregions are IBGE-defined statistical aggregations of municipalities. Results are robust to alternative window definitions; see Section 7.

The window dates should be interpreted as municipality-specific typical phenology rather than observed field-level phenology in each municipality-year. At the national scale, observed sowing, silking, and maturity dates are not available for every municipality and year. Large-scale maize studies therefore commonly anchor weather exposure to typical emergence, flowering, grain-fill, and maturity calendars, accepting this aggregation as the feasible way to align weather with crop physiology across broad geographies (Butts-Wilmsmeyer et al., 2019; Prudente Junior et al., 2024).

4.4 Heat and Water Deficit Measures

Within each phenological window w and muni-year (m, t), we construct heat exposure by adapting the GDD/KDD parameterization in Butler and Huybers (2013) and capping moderate heat accumulation at the 29°C damage threshold:

$$\mathrm{GDD}_{mwt} = \sum_{d \in w} \max\left(\min\left\{ \frac{T_{md}^{\max} + T_{md}^{\min}}{2},\ 29 \right\} - 8,\ 0 \right), \tag{8}$$

$$\mathrm{KDD}_{mwt} = \sum_{d \in w} \max(T_{md}^{\max} - 29,\ 0), \tag{9}$$

where $T_{md}^{\max}$ and $T_{md}^{\min}$ are daily maximum and minimum temperatures in °C. Growing degree days (GDD) capture the integral of moderate, growth-promoting heat between the maize base temperature of 8°C and the upper threshold of 29°C. On the other hand, killing degree days (KDD) capture cumulative exposure to extreme heat above the maize damage threshold of 29°C.

Water deficit is measured as the within-window sum of daily water deficit,

$$\mathrm{WD}_{mwt} = \sum_{d \in w} (\mathrm{ET}_{md} - P_{md}), \tag{10}$$

where ET_{md} is daily Penman-Monteith reference evapotranspiration and P_{md} is daily precipitation, both in millimeters. Positive values of water deficit indicate atmospheric demand exceeds rainfall (drought conditions), and negative values indicate the reverse (water surplus).

Unlike total precipitation, water deficit incorporates the demand side of the soil-moisture balance and therefore captures the joint role of heat and rainfall in determining crop water stress.

4.5 Sample Construction

We restrict the panel to observations with safrinha harvested area of at least 100 hectares, excluding observations with negligible safrinha production. We also restrict the sample to the seven most important second-season corn producing regions (Table 1), with MATOPIBA membership determined by the official municipality list (IBGE, 2022). After fixed-effect singleton removal, the estimation sample contains 17,565 municipality-year observations covering 1,370 municipalities over 2003–2023. The sample accounts for 95.9% of Brazil's second-season corn production over 2003–2023 and 95.4% over 2019–2023 (IBGE, 2024).

Table 1: Sample distribution by region

Region	**N**	**Municipalities**	**Production share (%)**
Mato Grosso (MT)	2,017	126	47.5
Paraná (PR)	6,761	355	15.4
Goiás (GO)	1,531	152	13.6
Mato Grosso do Sul (MS)	1,319	73	13.3
MATOPIBA	761	115	3.6
Minas Gerais (MG)	1,864	202	3.5
São Paulo (SP)	3,312	347	3.1
Total	17,565	1,370	100.0

Notes: MATOPIBA follows the official definition covering municipalities in Maranhão, Tocantins, Piauí, and Bahia (IBGE, 2022). The estimation sample excludes municipalities outside the seven main producing regions. Production share is each region's share of 2019–2023 mean second-season corn production within the sample, based on IBGE PAM (IBGE, 2024).

Mato Grosso accounts for 47.5% of recent safrinha production in the estimation sample, followed by Paraná (15.4%), Goiás (13.6%), and Mato Grosso do Sul (13.3%). With the exception of Paraná, which lies in the subtropical South, these three states are located in the midwestern Brazilian Cerrado. Three other regions complete the sample: MATOPIBA (3.6%), which encompasses municipalities in Maranhão (MA), Tocantins (TO), Piauí (PI), and Bahia (BA) (IBGE, 2022), and the states of Minas Gerais (3.5%) and São Paulo (3.1%). MATOPIBA is a relatively new frontier for second-season corn and has expanded rapidly over the sample period.

Table 2 reports summary statistics for the estimation sample. Safrinha yields average 4,312 kg/ha, with substantial variation across municipalities and years (standard deviation of 1,529 kg/ha). Over the four yield-cycle windows, the average municipality accumulates 2,007 growing degree days (GDD) and 197 killing degree days (KDD). Both heat exposures peak in the vegetative window, which alone accounts for 40% of cycle GDD and 51% of cycle KDD. The killing-degree share concentrates more sharply than the growing-degree share because the vegetative window also occupies the hottest portion of the safrinha cycle, in late summer, whereas subsequent windows progress into fall and winter as temperatures decline.

Water deficit follows a different seasonal pattern. The pre-planting and vegetative windows show a net surplus on average (means of -192 mm and -92 mm, respectively), consistent with their position at the tail of the wet season. From flowering onward, mean water deficit is close to balance: -5 mm in flowering, $+10$ mm in early grain-fill, and -6 mm in late grain-fill. Variation across municipality-years is large throughout, however, with standard deviations ranging from 73 mm in flowering to 223 mm in pre-planting, which shows evidence of substantial heterogeneity in moisture conditions across regions and years even within the same phenological window.

Table 2: Summary statistics, yield-regression sample

Variable	Mean	SD
Crop outcome		
Safrinha yield (kg/ha)	4,312	1,529
Growing degree days, GDD (8–29°C)		
Growing-season cycle (sum)	2,007	258
Vegetative	801	63
Flowering	456	55
Early grain-fill	428	83
Late grain-fill	322	83
Killing degree days, KDD		
Growing-season cycle (sum)	197	129
Vegetative	101	50
Flowering	46	35
Early grain-fill	30	38
Late grain-fill	20	34
Water deficit, ET − P (mm)		
Growing-season cycle (sum)	−93	206
Pre-planting	−192	223
Vegetative	−92	131
Flowering	−5	73
Early grain-fill	10	79
Late grain-fill	−6	84

Notes: The sample has 17,565 observations across 1,370 municipalities over 2003–2023 after applying the 100-hectare threshold, restricting the analysis to the seven main producing regions, and removing fixed-effect singletons. Water deficit is reference evapotranspiration minus precipitation, summed within window (positive = drought, negative = surplus). Window day boundaries (from planting day 0): pre-planting −90 to 0; vegetative 0–46; flowering 46–74; early grain-fill 74–105; late grain-fill 105–130. Growing-season cycle aggregates sum across the four yield-cycle windows (pre-planting excluded). Data are from IBGE PAM and BR-DWGD (IBGE, 2024; Xavier et al., 2022).

Figure 2 maps the municipalities in the analysis sample and the seven producing regions.

Six regions are state-based: Mato Grosso (MT), Goiás (GO), Minas Gerais (MG), Mato Grosso do Sul (MS), São Paulo (SP), and Paraná (PR). On the other hand, MATOPIBA is an official multi-state territorial classification containing 337 municipalities in Maranhão (MA), Tocantins (TO), Piauí (PI), and Bahia (BA) (IBGE, 2022).

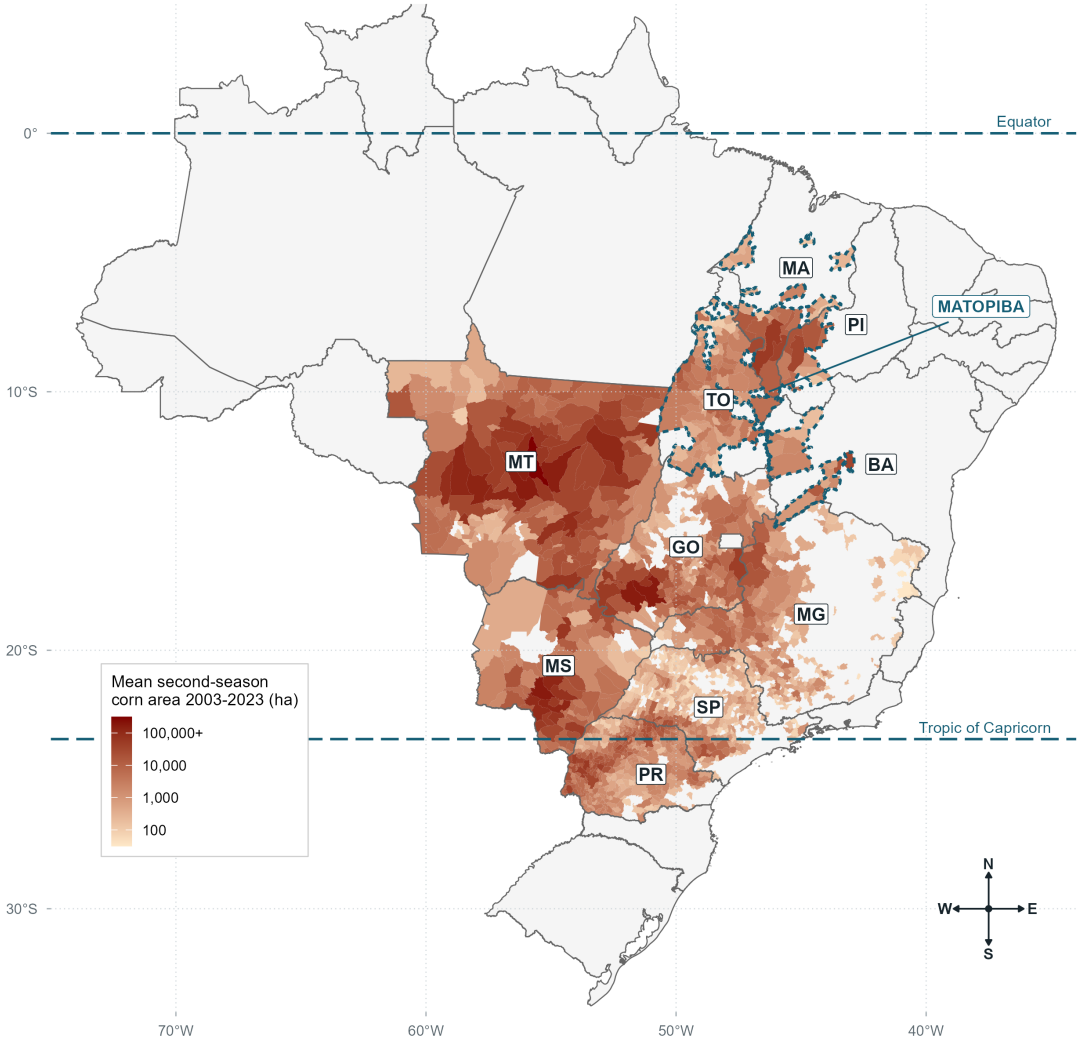


Figure 2: Spatial distribution of second-season corn area in the analysis sample.

Notes: Municipalities are shaded by mean second-season corn area over 2003–2023. Labels identify state abbreviations, including the four states containing official MATOPIBA municipalities. The dashed blue outline and callout identify the official MATOPIBA municipalities represented in the analysis sample. The Equator and Tropic of Capricorn are shown for geographic orientation. Source: authors' calculations using IBGE PAM and the IBGE MATOPIBA classification (IBGE, 2024, 2022).

5 Results

This section presents the estimated climate response of safrinha yields in two parts. We first estimate the national response on the full sample, for extreme heat and for water deficit. We then ask how the response varies across the major producing regions.

5.1 National Response

Figure 3 reports the estimated effect of one additional killing degree day (KDD, daily maximum temperature exceeding 29°C) on log safrinha yield. We compare two specifications. First, we pool heat exposure over the full growing season, the standard approach in the climate-econometrics literature. The marginal effect of KDD on yield in this specification is negative and statistically significant, with a point estimate of −0.09% per KDD (first bar).

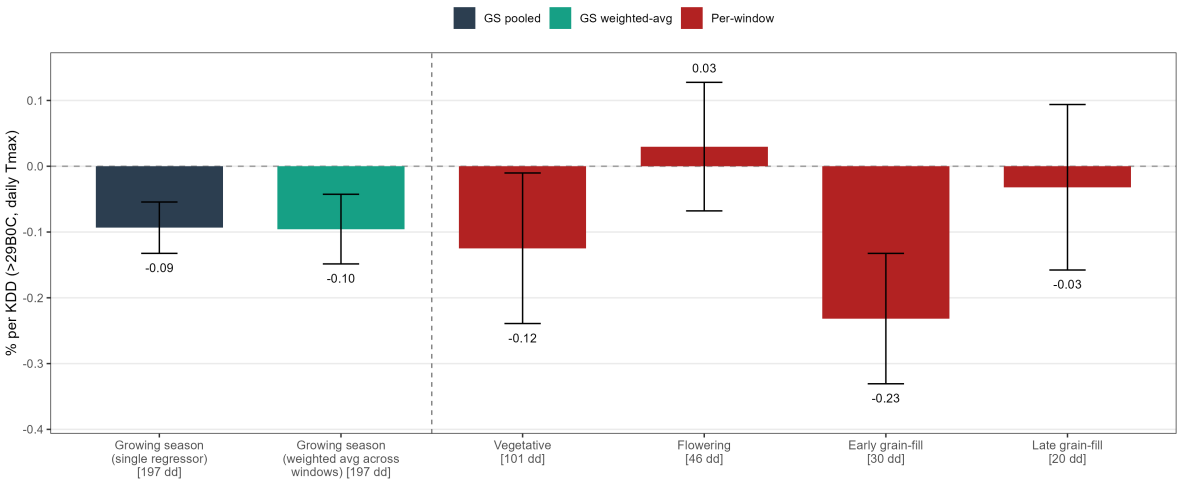


Figure 3: Extreme-heat damage: growing season vs phenological windows

Notes: Bars show the percentage change in log yield per killing degree day (KDD, > 29°C) from Equations 7 and 1; the weighted average is defined in Equation 2. Bracketed values show sample-mean KDD exposure. Error bars are 90% confidence intervals. Source: authors' estimates using the data described in Section 4.

The same data, decomposed across the four yield-cycle windows, shows that the aggregate heat effect is not evenly distributed through the crop cycle. Early grain-fill carries the largest per-KDD penalty (−0.23% per KDD), while the vegetative window also shows a negative effect (−0.13%) that is marginally significant. Flowering and late grain-fill are not statistically distinguishable from zero (+0.03% and −0.03%, respectively). The pattern is

consistent with the broader agronomic evidence that maize weather sensitivity depends on growth stage (Magalhães et al., 2002; Cicchino et al., 2010; Li et al., 2022; Andrea et al., 2019; Dong et al., 2021; Butts-Wilmsmeyer et al., 2019). It also accords with Butler and Huybers (2015), whose U.S. county-panel analysis finds KDD sensitivity generally largest during early grain filling, although the different production system and climate preclude a direct comparison with Brazilian second-season corn.

Because GDD and KDD are constructed from the same daily temperature distribution, these KDD coefficients identify extreme-heat effects conditional on moderate heat accumulation, a qualification that is especially important for flowering. The +0.03% flowering estimate is statistically indistinguishable from zero and becomes negative in otherwise identical specifications that omit GDD (Appendix Table A3), while the growing-season pooled KDD estimate is essentially unchanged and the window-weighted estimate remains negative but smaller in magnitude. We therefore do not interpret the flowering coefficient as evidence that extreme heat is beneficial; the robust heat result is aggregate damage, with the most stable window-specific penalty appearing during early grain-fill.

The weighted-average coefficient across windows (second bar), which aggregates the per-window effects back to the growing-season scale, is −0.10% per KDD. It is statistically significant and nearly identical to the pooled estimate. Thus, phenological decomposition does not substantially change the aggregate national heat effect. We use the window-weighted estimate as the main aggregate heat effect because it is consistent with the phenological specification used throughout the paper. The value of phenological windows is to show which parts of the crop cycle contribute to that aggregate damage. This highlights the importance of splitting the growing season into phenological windows when interpreting mechanisms and regional exposure, even when the pooled growing-season coefficient is close to the window-weighted average (Ortiz-Bobea, 2013; Wen et al., 2026).

Using the window-weighted estimate and the sample-mean exposure of 197 KDD, the implied effect on log yield at typical conditions is about −20% relative to a zero-KDD

counterfactual. Because the dependent variable is log yield, the exact level conversion is $\Delta y/y = \exp(\Delta \log y) - 1$. Observed yields are therefore roughly 18% below what they would be at zero extreme heat exposure, while yields at zero extreme heat would be about 22% above observed yields. At sample-mean exposure, the vegetative window contributes about -13% of log yield ($101 \times -0.13\%$) because it receives the largest volume of extreme heat, while early grain-fill contributes about -7% ($30 \times -0.23\%$) because it carries the largest per-KDD penalty. Flowering contributes little at the sample mean because its coefficient is close to zero, and late grain-fill contributes less than one log point.

We now turn to the water-deficit response. Water stress is potentially very damaging for second-season corn, because the crop is sown as the rainy season begins to recede, so its most water-sensitive phases, flowering and grain-fill, coincide with declining rainfall (Andrea et al., 2019; Prudente Junior et al., 2024). Figure 4 plots the estimated marginal effect of water deficit (evapotranspiration minus precipitation, so that positive values denote drier conditions) on log safrinha yield. Because water deficit enters the regression quadratically, we plot the marginal effect across the observed range of deficit rather than a single slope.

Panel (a) compares two representations of the same growing-season response, a single coefficient estimated on total cycle deficit (growing-season pooled) and the response built up from the four yield-cycle window coefficients (window-aggregated). The two curves have a similar shape over the observed range: additional deficit is more damaging as the total cycle becomes drier. At the sample mean total-cycle water deficit (-93 mm, a net surplus), the window-aggregated marginal effect is about -0.03% per mm, and the implied total effect is $+2.35\%$ in log yield relative to a zero-deficit counterfactual. The corresponding growing-season-pooled total effect is $+2.40\%$. These positive total effects reflect average surplus conditions; moving from water surplus toward deficit lowers safrinha yields.

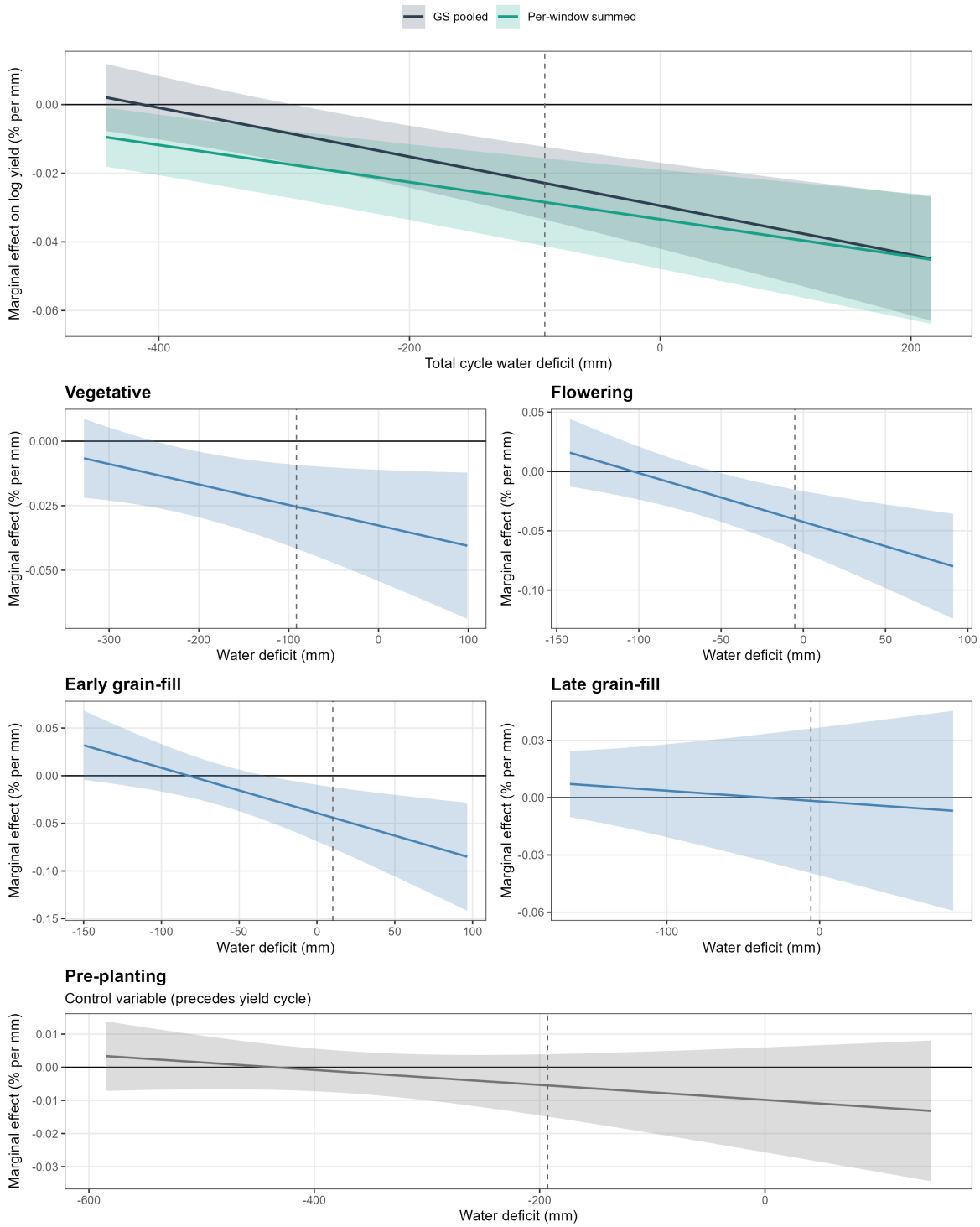


Figure 4: Water-deficit response: growing season vs phenological windows

Notes: Curves show the marginal effect of water deficit on log yield across the 5th–95th percentile range. Panel (a) overlays the growing-season-pooled specification (Equation 7) and the window-aggregated response (Equation 1), which excludes pre-planting. The other panels show each yield-cycle window and the pre-planting control. Positive values denote drier conditions. Dashed vertical lines mark sample means; shaded bands are 90% confidence intervals. Source: authors' estimates using the data described in Section 4.

Decomposed across windows, water deficit becomes most damaging as the reproductive windows dry. The vegetative window is modestly negative but imprecisely estimated across much of its range. Flowering and early grain-fill both show the clearest biological pattern: when these windows are wet, additional water deficit is not clearly damaging, but as the windows cross into drier conditions the marginal effect turns negative. Late grain-fill is close to zero and imprecise in the estimation sample. We therefore interpret the water-deficit evidence as strongest for the flowering and early grain-fill portions of the crop cycle, rather than as evidence of a precisely estimated late-cycle response.

The positive or near-zero marginal effect in flowering and early grain-fill at the wettest conditions indicates that additional water deficit is not harmful when the window remains in moisture surplus. The relationship turns damaging once conditions cross toward deficit. The pre-planting control is small and imprecise across its range, indicating that moisture accumulated before the cycle adds little once the within-cycle windows are conditioned on.

Comparing the heat and water-deficit effects at the national level, the main value of the phenological decomposition is interpretive. For heat, the pooled and window-weighted aggregate estimates are similar, but the window estimates show that exposure volume in the vegetative stage and per-KDD sensitivity in early grain-fill jointly drive the national burden. For water deficit, the pooled and window-aggregated responses also broadly agree in the aggregate, while the window decomposition shows that flowering and early grain-fill are the most interpretable drought-sensitive stages. This agrees with agronomic evidence that water availability during flowering and grain-fill is a key limiter of maize yield (Butts-Wilmsmeyer et al., 2019; Prudente Junior et al., 2024).

Appendix Table A1 reports the underlying national growing-season and phenological regression estimates.

5.2 Regional Heterogeneity

The regional-interaction model is used to characterize spatial heterogeneity and to assess whether the national result masks offsetting regional patterns. The safrinha system spans a

wide climatic gradient, from subtropical Paraná to the tropical Cerrado and the MATOPIBA frontier (see Section 2). A single national coefficient masks regional differences that matter for understanding which producing areas are most exposed and how the system may evolve under warming. We estimate Equation 1 with region interactions for each per-window coefficient, and examine both channels in turn. We begin with heat, and report the region-specific weighted-average marginal heat damage in Figure 5.[3] We use the window-weighted-average approach because it captures the per-window heterogeneity that the growing-season-pooled coefficient averages over.

The marginal effect of KDD on yield varies sharply across regions. Goiás (GO) carries the largest per-day damage (-0.20%), followed by Paraná (PR) (-0.14%), Minas Gerais (MG) (-0.13%), and Mato Grosso do Sul (MS) (-0.11%). MATOPIBA shows a marginal effect of -0.10% per KDD but is not significant at conventional levels. São Paulo (SP) is also small and statistically indistinguishable from zero (-0.03%). Mato Grosso (MT), the dominant safrinha producer, has a near-zero point estimate ($+0.01\%$) and a confidence interval that comfortably spans zero.

[3] Regions are defined in Table 1.

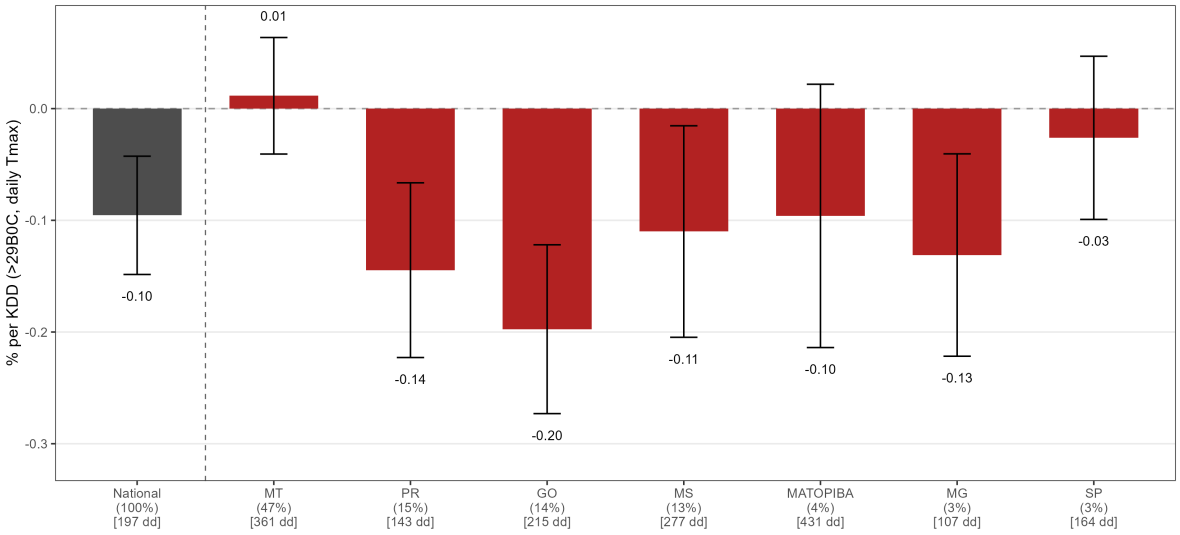


Figure 5: Extreme-heat damage by region

Notes: Bars show the weighted-average marginal heat damage per killing degree day (KDD, > 29°C) from Equations 1 and 2, nationally and by region. Parentheses show production shares; brackets show sample-mean KDD exposure. Regions are ordered by production share. Error bars are 90% confidence intervals. Source: authors' estimates using the data described in Section 4.

The Mato Grosso estimate should be interpreted cautiously. MT accounts for 47% of safrinha production in the estimation sample and is among the most heat-exposed regions, with 361 KDD on average over the growing season, yet its conditional KDD response is not statistically distinguishable from zero. A diagnostic specification that removes GDD makes the KDD slope more negative. The near-zero estimate therefore should not be read as evidence that MT is biologically insensitive to heat. Section 8 returns to this interpretation.

Similar to the approach used for the heat results, we test whether the water-deficit effect varies by region, given the substantial climatic differences between producing areas. Figure 6 shows the per-region marginal effect of water deficit, computed from the per-window decomposition. The National panel is the national window-aggregated response and reproduces the national finding.

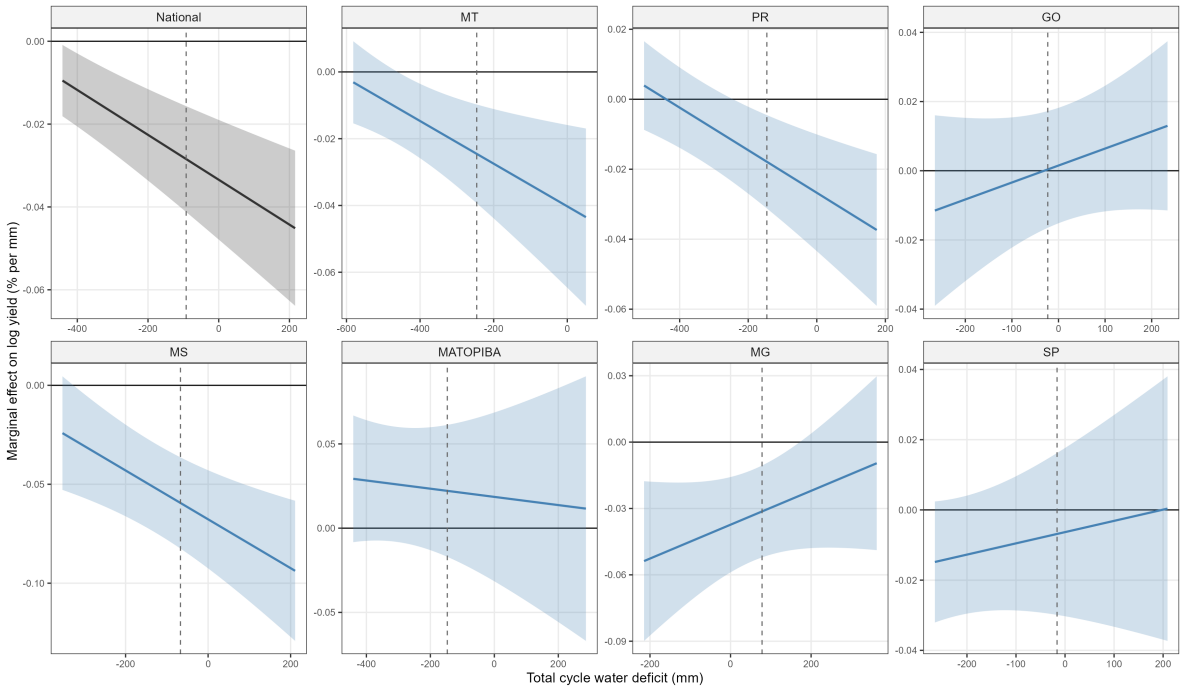


Figure 6: Water-deficit response by region

Notes: Curves show the marginal effect of water deficit on log yield across the 5th–95th percentile range, nationally and by region. The marginal effect is computed from the per-window decomposition in Equation 1. Positive values denote drier conditions. Shaded bands are 90% confidence intervals. Source: authors' estimates using the data described in Section 4.

The regional water-deficit curves show that the water channel does not map one-for-one onto the heat channel. Paraná (PR), Mato Grosso do Sul (MS), and Mato Grosso (MT) track the national pattern most clearly: the marginal effect becomes more negative as total-cycle deficit increases, and the estimates are distinguishable from zero over the drier portion of each region's distribution. MATOPIBA is imprecisely estimated throughout.

The remaining regions show weaker or less monotonic water-deficit patterns. Minas Gerais (MG) is negative over much of the range but becomes less precise in the dry tail. Goiás (GO) and São Paulo (SP) have confidence bands that cover zero across much of the range, so their water-deficit curves should not be read as strong evidence of either damage or protection. The regional decomposition therefore shows that exposure to water deficit, like exposure to heat, is uneven across producing regions, and the two channels do not line up region by region. GO, MG, PR, and MS carry clear heat penalties, while PR, MS, and

MT show the clearest water-deficit losses. MT's heat response is muted, but its water-deficit response remains an important source of risk.

Appendix Table A2 summarizes the regional growing-season and window-weighted heat estimates together with water-deficit marginal effects at region-specific distribution points.

6 Uniform Warming Counterfactuals

After estimating the national and regional weather-response functions, we translate the main national phenological specification into yield effects under uniform warming counterfactuals. This exercise is not a full climate projection: it isolates the implications of warmer daily temperature distributions, holding precipitation fixed except for the ET0-induced water-deficit channel. For this exercise, we add 1°C, 2°C, and 3°C to the daily temperature panel, recompute GDD and KDD in each phenological window, and update reference evapotranspiration while holding precipitation fixed. Appendix B describes the construction in detail. For national interpretation, we use the main national phenological specification because it directly estimates the average response for the full sample while preserving the per-window structure used throughout the paper.

Table 3: Yield effects under uniform warming counterfactuals

	+1°C	+2°C	+3°C
Main national phenological specification			
Heat channel, GDD + KDD	−7.5	−15.2	−22.8
Water-deficit channel	−0.7	−1.4	−2.2
Net effect	−8.2 [−12.5, −3.6]	−16.4 [−24.3, −7.8]	−24.4 [−35.0, −12.4]

Notes: Entries are production-weighted yield-level percentage changes, computed as $100 \times [\exp(\Delta \log y) - 1]$. Brackets report 90% coefficient-simulation intervals from 10,000 draws using the cluster-robust covariance matrix, following Gammans et al. (2025). Each counterfactual uniformly increases daily temperature, recomputes GDD, KDD, and ET0, and holds precipitation fixed. Intervals capture coefficient uncertainty only; Appendix B details the construction and maintained assumptions. Source: authors' estimates using the data described in Section 4.

Table 3 shows that the implied losses are economically large. Under the main national phenological specification, a 1°C warming reduces yield levels by 8.2%, with a 90% coefficient-simulation interval from −12.5% to −3.6%. The implied losses rise to 16.4% under 2°C and 24.4% under 3°C, and the coefficient-simulation intervals remain below zero in all three scenarios. Most of the implied damage comes from the heat channel, while the water-deficit channel adds smaller additional losses through higher reference evapotranspiration.

The central national message is that even when the growing-season-pooled and phenologically weighted marginal KDD estimates are similar, translating the response into warmer daily temperature distributions implies substantial yield losses for second-season corn.

7 Robustness Checks

This section shows the robustness of the main results to several subsamples and specifications. First, we analyze the effect of removing irrigated municipalities from the sample. We then summarize how the main estimates respond to dropping individual regions, weighting the regression, using alternative fixed effects, and using different window definitions.

7.1 Irrigation

We test the heat and water-deficit results after removing municipalities with non-trivial irrigation. Using the agricultural census, we measure irrigated area relative to temporary-crop area and drop municipalities above 10% in either 2006 or 2017. The 10% threshold is motivated by Butler and Huybers (2015), who use the same cutoff to exclude heavily irrigated counties from a panel study of U.S. maize, although their denominator is total harvested area. We also apply a stricter 5% cutoff as a sensitivity check. These thresholds are screening rules rather than natural boundaries between rainfed and irrigated production, and reporting both shows whether the results depend on the chosen definition.

Figure 7 compares the heat coefficients across irrigation samples, for the growing season as a whole and by phenological window. The aggregate heat effect is unchanged. Both the pooled and the window-aggregated growing-season estimates are essentially identical across the three samples, so the main heat damage estimate is not an artifact of including irrigated municipalities. The per-window decomposition is also stable in its main feature: early grain-fill remains the largest and most precisely estimated penalty, while vegetative exposure is negative and flowering and late grain-fill are less precisely estimated.

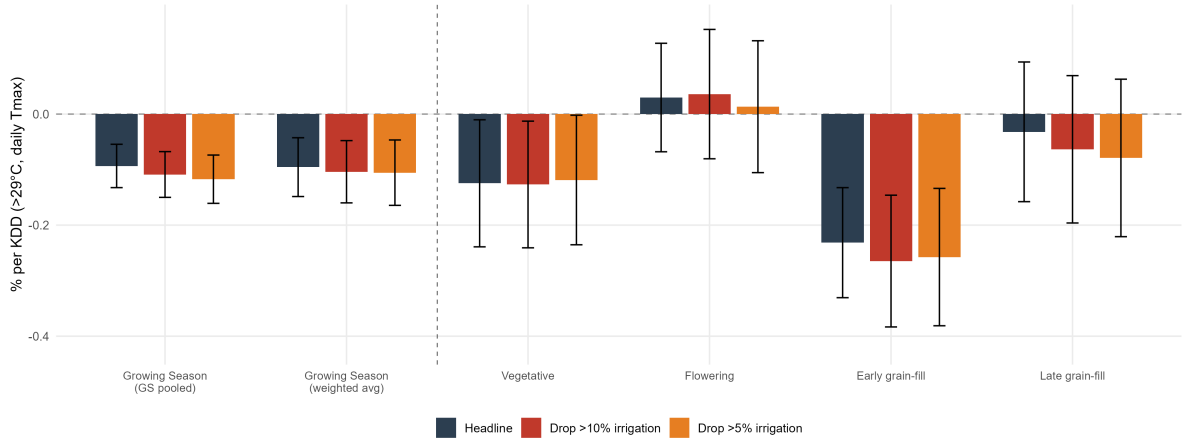


Figure 7: Heat response: growing season vs. phenological windows, by irrigation status

Notes: Bars show the percentage change in log yield per killing degree day (KDD, > 29°C) from Equations 7 and 1. The main sample is the full main-results sample; the other bars exclude municipalities above the 10% or 5% irrigation thresholds. Error bars are 90% confidence intervals. Source: authors' estimates using the data described in Section 4 and IBGE Agricultural Census irrigation measures.

Figure 8 shows the water-deficit response by irrigation status, for the growing season and by phenological window. The growing-season curves overlap across the common range of deficit, so dropping irrigated municipalities does not change the qualitative water-deficit response. At the sample mean, the irrigation-filtered marginal effects are somewhat less negative than the main estimate, but remain negative. Across phenological windows, flowering and early grain-fill continue to be the windows where the response turns most damaging as conditions dry. Overall, the water-deficit response is similar across the main and irrigation-dropped samples.

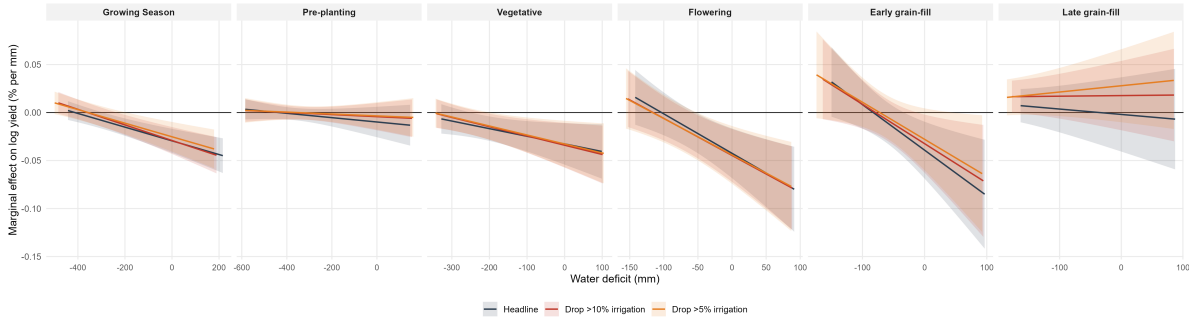


Figure 8: Water-deficit response: growing season vs. phenological windows, by irrigation status

Notes: Curves show marginal water-deficit effects across the 5th–95th percentile range. The leftmost panel is the growing-season-pooled specification; the remaining panels are the pre-planting control and four yield-cycle windows from Equation 1. Curves overlay the main sample and the two irrigation-dropped samples. Positive values denote drier conditions. Shaded bands are 90% confidence intervals. Source: authors' estimates using the data described in Section 4 and IBGE Agricultural Census irrigation measures.

7.2 Overview of robustness checks

Beyond irrigation, we re-estimate the main specification under a series of alternative choices, changing one feature at a time and holding everything else at the main setting. The variants fall into five groups. *Sample restriction* drops one producing region at a time, to check that no single region drives the result. *Weighting* re-estimates the regression weighted by each municipality's mean safrinha production (2019–2023), so that large producers count more, rather than the unweighted main estimate. *FE alternative* replaces the main fixed effects (municipality, agricultural year, and region-specific linear trends) with the simpler municipality and state-by-year structure used as a robustness check. *Window structure*

replaces the four yield-cycle windows with a coarser three-window definition (vegetative, flowering, grain-fill). *Flowering anchor* shifts the assumed silking date by plus or minus seven days, which moves the boundaries of the vegetative and flowering windows while holding the planting-to-maturity length fixed, to check that the results do not depend on the precise dating of the reproductive window.

We summarize each variant by a single marginal effect at mean conditions. Because heat enters the yield equation linearly while water deficit enters quadratically, the two channels are summarized by different aggregates. For heat we report the weighted-average marginal effect, the four per-window extreme-heat coefficients averaged by their mean exposure. For water deficit we report the marginal effect of the window-aggregated response evaluated at the sample-mean deficit. Figures 9 and 10 report these for every variant, with the main estimate shown as a reference line.

The heat effect is stable across every variant. The weighted-average marginal effect moves little and the confidence bands overlap the main estimate throughout. The two variants that move it both involve removing an entire region from the set of producing regions. Dropping Paraná widens the band until it reaches zero, and dropping São Paulo shifts the point more negative with a wide band. Neither overturns the main estimate. The effect holds under production weighting, the alternative fixed effects, the coarser window definition, and the shifted flowering anchor, so it is not an artifact of a particular sample, weighting scheme, fixed-effect structure, or window definition.

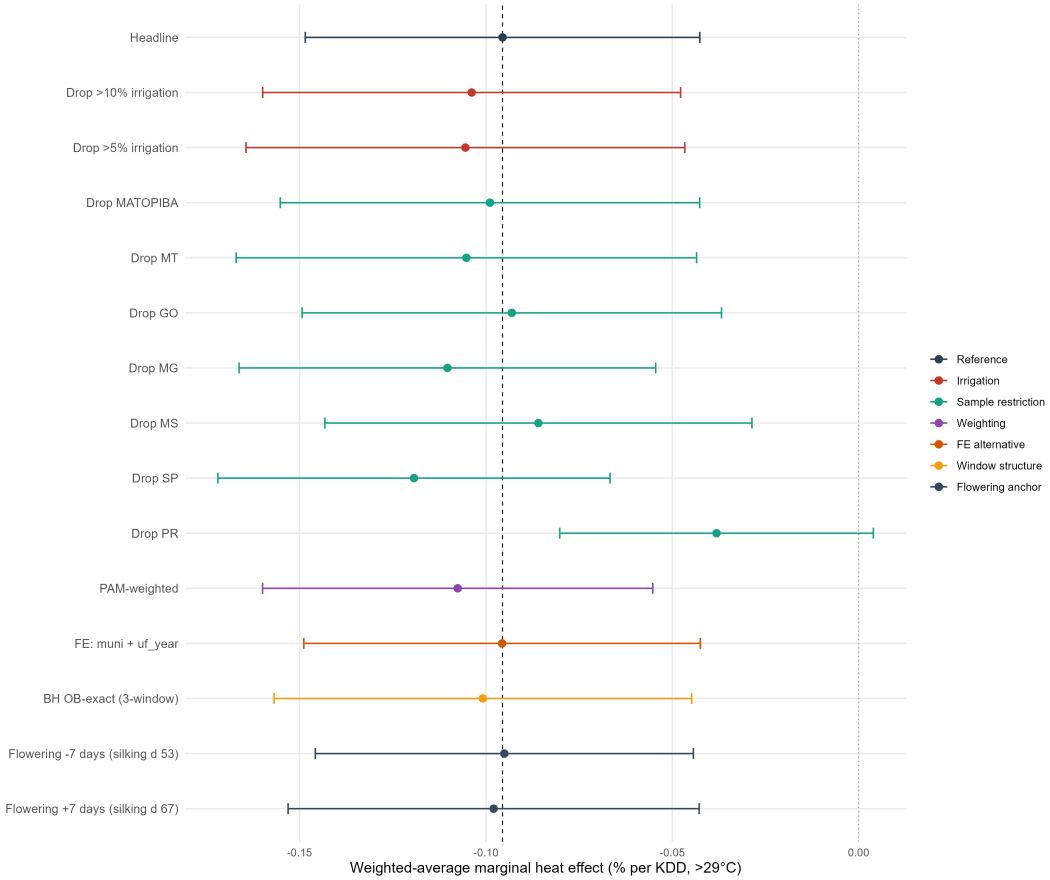


Figure 9: Robustness of the weighted-average marginal heat effect

Notes: Each row reports the weighted-average marginal extreme-heat effect, $\bar{\beta}^{\mathrm{KDD}} = \sum_w \overline{\mathrm{KDD}}_w \, \beta_w^{\mathrm{KDD}} \big/ \sum_w \overline{\mathrm{KDD}}_w$, the exposure-weighted average of the four per-window extreme-heat coefficients β_w^{KDD} from Equation 1, for one robustness variant. Rows are colored by variant group. The dashed vertical line is the main estimate (full sample, baseline specification). Horizontal bars are 90% confidence intervals (delta method). Source: authors' estimates using the data described in Section 4.

The water-deficit effect is also stable in sign across the same alternatives. All single-region-drop variants remain negative and statistically distinguishable from zero, indicating that the national water-deficit result is not driven by one producing region. The irrigation-filtered estimates are somewhat attenuated but remain negative. Production weighting makes the effect more negative, while the coarser window structure and the flowering-anchor shifts remain close to the main estimate. The main sensitivity is the fixed-effect alternative: replacing the main municipality, year, and region-trend structure with municipality and state-by-year fixed effects attenuates the water-deficit effect and produces a confidence interval that crosses zero. This is consistent with the state-by-year structure absorbing a

larger share of the within-state climate variation that identifies the water-deficit response.

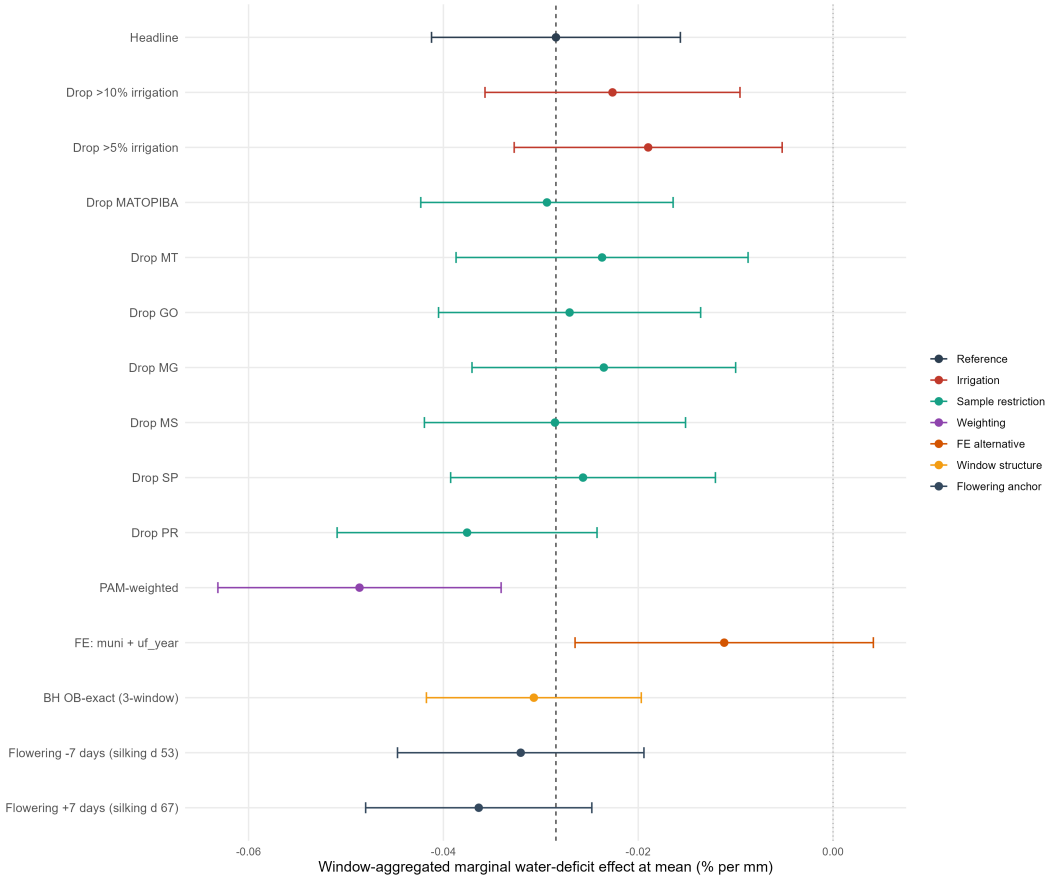


Figure 10: Robustness of the window-aggregated marginal water-deficit effect

Notes: Each row reports the window-aggregated marginal water-deficit effect at the sample-mean deficit, $\sum_w \omega_w \left(\gamma_w + 2\delta_w \overline{\mathrm{WD}}_w\right)$, where γ_w and δ_w are the per-window linear and quadratic water-deficit coefficients from Equation 1, ω_w is the slope from regressing window-w deficit on total-cycle deficit ($\sum_w \omega_w = 1$), and $\overline{\mathrm{WD}}_w$ is the window-w mean deficit. The pre-planting window is included in the regression as a control but excluded from this summary. Rows are colored by variant group. The dashed vertical line is the main estimate. Horizontal bars are 90% confidence intervals (delta method). Source: authors' estimates using the data described in Section 4.

Together, the robustness checks support two conclusions. First, the heat result is highly stable: the main effect remains close to -0.10% per KDD under irrigation filters, individual-region drops, production weighting, alternative fixed effects, and alternative window definitions. Second, the water-deficit result is negative in nearly all variants, but more sensitive to fixed-effect choice than the heat result. We therefore treat the heat estimate as the more stable of the two main marginal effects, while interpreting the water-deficit estimate as robust in sign but more dependent on the identifying variation retained by the fixed-effect

structure.

8 Discussion

Brazil's second-season corn matters well beyond Brazil because the country is now the world's third-largest corn producer and second-largest exporter, following three decades in which corn output more than tripled while planted area rose by roughly one-half (USDA Foreign Agricultural Service, 2026; Moreira Guimarães et al., 2022), and this position increasingly rests on a second season that supplies roughly 80% of national output (CONAB, 2025). Its transition from a marginal, low-input crop to Brazil's dominant corn season forms part of the agricultural transformation often described as the "Miracle of the Cerrado," in which tropical germplasm, soil correction, no-till planting, machinery, crop protection, and soybean-corn succession made two commercial harvests from the same land possible within one agricultural year (Jeddi and DePaula, 2026; Moreira Guimarães et al., 2022). Although second-season corn remains predominantly rainfed (Prudente Junior et al., 2024), it enters the field after the soybean harvest and reaches critical stages as the rainy season recedes, so the compressed calendar that greatly expanded grain supply also concentrates climate risk in the season that is now central to Brazilian corn production. Given Brazil's role in trade, shocks to this crop can therefore affect grain availability and feed markets beyond the country.

The estimates show why climate change threatens this achievement because, unlike temperate double-cropping systems in which warmer conditions can relax thermal constraints on planting a second crop (Gammans et al., 2025), Brazil's second-season corn already occupies a thermally feasible tropical and subtropical window. Additional warming is therefore more likely to shift exposure toward extreme heat, raise reference evapotranspiration, and intensify water stress as the rainy season retreats than to enlarge the feasible production window, which accords with crop-modeling evidence that changes in rainfall and wet-season timing pose additional risks to soybean-corn systems in central and northern Brazil (Pires et al., 2016; Hampf et al., 2020; Bigolin and Talamini, 2024).

Brazil's agricultural history also shows why adaptation involves complementary public and private roles. Embrapa contributed historically to the tropical maize germplasm base and its adaptation to acidic Cerrado soils; its BR201 hybrid reached 12% of the national maize market in the early 1990s, and germplasm developed by the institution was subsequently distributed to public institutes and private seed companies (Moreira Guimarães et al., 2022). In a recent nonrandom survey of digitally managed farms in six major second-season corn states, the leading hybrids were supplied by private companies, and three corporate groups accounted for 65.2% of the mapped area (Aegro Insights, 2026). Adaptation is therefore likely to depend on complementarity between private breeding of regionally appropriate hybrids and public research, testing, and extension on heat and drought response, planting calendars, and soil-water management.

Because second-season corn is predominantly rainfed, irrigation should be understood as a geographically limited adaptation option rather than a description of current national practice. Where water and infrastructure are available, our estimates suggest that supplemental irrigation would have greatest value around flowering and early grain fill. Any expansion would confront resource constraints because Brazil's National Water and Sanitation Agency projects substantial growth in irrigated area through 2040 and emphasizes planning around water security and competing uses (Agência Nacional de Águas e Saneamento Básico, 2021). Western Bahia illustrates this tension because groundwater from the Urucuia Aquifer System supports irrigation while aquifer discharge sustains regional streamflow; recent evidence identifies pumping as the principal driver of declining low flows and shows that climate variability can intensify low-flow regimes and water demand (Rodrigues et al., 2024). Groundwater can therefore buffer rainfall shocks only within hydrological limits, requiring region- and stage-targeted irrigation under integrated basin management that accounts for capital and energy costs, water rights, aquifer recharge, environmental flows, and competing users.

The regional estimates reinforce the need for place-specific adaptation. One source of regional heterogeneity is hybrid choice. A recent AEGRO Insights survey documents distinct

hybrid portfolios by state and greater use of earlier-maturity materials in Paraná than in the Cerrado states, consistent with the South's shorter production window (Aegro Insights, 2026).

Several limitations define the scope of these implications because the phenological calendars represent typical municipality-specific timing rather than annual field-level planting and silking dates; although the estimates are robust to shifting the flowering anchor by seven days, remaining timing error may attenuate differences across windows. Water deficit, measured as reference evapotranspiration minus precipitation, captures atmospheric demand relative to rainfall rather than field-level soil moisture and cannot account directly for soil-water storage, rooting depth, or irrigation at individual farms.

9 Conclusion

This paper estimates the weather response of second-season corn in Brazil's major producing regions using municipality-year variation in daily heat and water deficit. The sample covers the core of the national safrinha system, accounting for 95.9% of Brazil's second-season corn production over 2003–2023 and 95.4% over 2019–2023. The results show that extreme heat is damaging to yields, with a main window-weighted marginal effect close to -0.10% per killing degree day. When the main national phenological specification is applied to uniform warming counterfactuals, a 1°C warming reduces predicted yield levels by 8.2%, with losses rising to 16.4% under 2°C and 24.4% under 3°C.

Our results indicate that the climate vulnerability of Brazil's second-season corn is concentrated within specific parts of its compressed production cycle. Heat effects are concentrated in early grain fill, while water-deficit effects are strongest during flowering and early grain fill. Regional estimates show the largest negative conditional heat responses in Goiás, Minas Gerais, Paraná, and Mato Grosso do Sul, while estimates for Mato Grosso, São Paulo, and MATOPIBA are smaller or less precise. These patterns point to adaptation priorities in timely planting, hybrid choice, soil-moisture conservation, and supplemental irrigation.

Appendix

A Main Regression Results

Table A1: National climate-response estimates

	Growing season	Vegetative	Flowering	Early grain-fill	Late grain-fill
Heat (percent log-yield change per degree day)					
GDD	−0.004	+0.126*	−0.084	−0.032	−0.022
	(0.020)	(0.067)	(0.061)	(0.039)	(0.037)
KDD (> 29°C)	−0.093***	−0.125*	+0.030	−0.232***	−0.032
	(0.024)	(0.070)	(0.059)	(0.060)	(0.076)
Window-weighted KDD	–			−0.096***	
				(0.032)	
Water deficit					
Linear (percent per mm)	−0.029***	−0.033**	−0.042***	−0.039**	−0.002
	(0.008)	(0.013)	(0.016)	(0.018)	(0.024)
Quadratic ($\times 10^6$)	−0.357***	−0.397	−2.054***	−2.373**	−0.280
	(0.095)	(0.242)	(0.775)	(1.008)	(0.503)
Pre-planting water-deficit controls					
Linear (percent per mm)	−0.009			−0.010	
	(0.010)			(0.010)	
Quadratic ($\times 10^6$)	−0.120			−0.113	
	(0.126)			(0.119)	
Municipality fixed effects	Yes			Yes	
Agricultural-year fixed effects	Yes			Yes	
Region-specific linear trends	Yes			Yes	
Within R^2	0.038			0.053	
Observations	17,565			17,565	

Notes: The dependent variable is log second-season corn yield. The growing-season column reports Equation 7; the four window columns report one phenological regression, Equation 1. The window-weighted KDD estimate aggregates the four window coefficients using mean KDD exposure. Standard errors, clustered by state-year, are in parentheses. All models include the fixed effects and trends shown. Source: authors' estimates using the data described in Section 4.

Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Regional climate-response estimates

	Heat effect per KDD (percent)		Water-deficit marginal effect (percent per mm)		
Region	Growing season	Window-weighted	P25	P50	P75
National	−0.09	−0.10	−0.02	−0.03	−0.04
	[-0.13, -0.05]	[-0.15, -0.04]	[-0.03, -0.01]	[-0.04, -0.02]	[-0.05, -0.02]
MATOPIBA	−0.13	−0.10	+0.03	+0.02	+0.02
	[-0.28, +0.02]	[-0.21, +0.02]	[-0.01, +0.06]	[-0.01, +0.06]	[-0.03, +0.07]
MT	+0.01	+0.01	−0.02	−0.03	−0.03
	[-0.03, +0.05]	[-0.04, +0.06]	[-0.03, -0.01]	[-0.04, -0.01]	[-0.05, -0.01]
GO	−0.23	−0.20	−0.01	+0.00	+0.01
	[-0.31, -0.15]	[-0.27, -0.12]	[-0.03, +0.02]	[-0.02, +0.02]	[-0.01, +0.02]
MG	−0.14	−0.13	−0.04	−0.03	−0.02
	[-0.22, -0.06]	[-0.22, -0.04]	[-0.06, -0.02]	[-0.05, -0.01]	[-0.05, +0.00]
MS	−0.03	−0.11	−0.04	−0.06	−0.07
	[-0.09, +0.04]	[-0.20, -0.02]	[-0.07, -0.02]	[-0.08, -0.04]	[-0.10, -0.05]
SP	+0.06	−0.03	−0.01	−0.01	−0.00
	[+0.01, +0.11]	[-0.10, +0.05]	[-0.03, +0.01]	[-0.03, +0.02]	[-0.03, +0.03]
PR	−0.16	−0.14	−0.01	−0.02	−0.03
	[-0.23, -0.10]	[-0.22, -0.07]	[-0.02, +0.00]	[-0.03, -0.00]	[-0.04, -0.01]
Within R^2	0.066	0.115		0.115	
Observations	17,565	17,565		17,565	

Notes: Brackets contain 90% confidence intervals. Heat entries are percent log-yield changes per additional KDD. Window-weighted estimates aggregate the four region-specific phenological coefficients using regional mean KDD exposure. Water-deficit entries are marginal effects evaluated at each region's own 25th, 50th, and 75th percentiles of total-cycle deficit, from wetter to drier conditions. The National row uses the corresponding national models. All models include municipality and agricultural-year fixed effects and region-specific linear trends, with standard errors clustered by state-year. Source: authors' estimates using the data described in Section 4.

B Uniform Warming Counterfactual Construction

The warming exercise is a uniform-shift counterfactual designed to translate the estimated weather-response functions into yield-level effects under warmer daily temperature distributions. It should be read as a sensitivity exercise, not as a full climate projection with modeled changes in rainfall, technology, planting dates, or other adaptation margins. Starting from

the estimation-sample municipality-day weather panel, we add 1°C, 2°C, and 3°C to daily minimum and maximum temperature. For each municipality-year-window observation, we then recompute GDD and KDD from the shifted daily temperatures using the same 8°C lower threshold and 29°C upper threshold used in the regressions. Let $\Delta GDD_{iwt}(s)$ and $\Delta KDD_{iwt}(s)$ denote the difference between shifted and baseline exposure for municipality i, window w, year t, and warming scenario $s \in \{1, 2, 3\}$. The heat-channel log-yield change is

$$\Delta \log y^{H}_{iwt}(s) = \hat{\beta}^{GDD}_{w} \Delta GDD_{iwt}(s) + \hat{\beta}^{KDD}_{w} \Delta KDD_{iwt}(s). \tag{11}$$

The water-deficit channel updates reference evapotranspiration while holding precipitation fixed. The stored warming routine estimates an ET0 temperature elasticity of 0.0516 per 1°C. For a warming scenario s, baseline reference evapotranspiration in each municipality-year-window is multiplied by $\exp(0.0516s) - 1$, giving proportional ET0 increases of 5.3%, 10.9%, and 16.7% for 1°C, 2°C, and 3°C, respectively. Because precipitation is fixed, this ET0 increase is also the change in water deficit:

$$\Delta WD_{iwt}(s) = ET0_{iwt}\{\exp(0.0516s) - 1\}. \tag{12}$$

The water-deficit effect uses the linear and quadratic coefficients from the same phenological specification:

$$\Delta \log y^{W}_{iwt}(s) = \hat{\theta}_{w} \Delta WD_{iwt}(s) + \hat{\phi}_{w} \left[(WD_{iwt} + \Delta WD_{iwt}(s))^2 - WD^2_{iwt} \right]. \tag{13}$$

We sum the heat and water-deficit components within each scenario and aggregate the resulting municipality-year-window predictions using recent production weights. Specifically, the weights are based on each municipality's mean second-season corn production over 2019–2023. Within each phenological window, these production weights are normalized over the estimation-sample municipality-years, and the national prediction sums the weighted com-

ponent changes across windows. For the main national counterfactual, we apply the national phenological coefficients to each municipality-year-window observation before aggregation. For the sensitivity check, we repeat the same calculation using the region-specific coefficients from the regional-interaction model and aggregate the resulting predictions with the same production weights. The table reports yield-level percentage changes, computed as $100 \times [\exp(\Delta \log y) - 1]$.

For statistical uncertainty, we follow the coefficient-simulation approach used by Gammans et al. (2025). We draw 10,000 coefficient vectors from the estimated coefficient vector and cluster-robust variance-covariance matrix, holding the weather perturbations, ET0 elasticity, precipitation, production weights, planting dates, technology, and adaptation margins fixed. For each draw, we recompute the heat, water-deficit, and net warming effects using the same aggregation described above. The bracketed intervals in Table 3 are the 5th and 95th percentiles of the resulting yield-level net-effect distributions.

C GDD–KDD Specification Sensitivity

Table A3: Sensitivity of national KDD estimates to omitting GDD

Estimand	With GDD	No GDD	Difference
Growing-season pooled	-0.093 [-0.132, -0.054]	-0.095 [-0.136, -0.055]	-0.002
Window-weighted average	-0.096 [-0.148, -0.043]	-0.070 [-0.111, -0.030]	+0.025
Vegetative	-0.125 [-0.239, -0.010]	-0.032 [-0.104, +0.039]	+0.092
Flowering	+0.030 [-0.068, +0.128]	-0.046 [-0.124, +0.031]	-0.076
Early grain-fill	-0.232 [-0.331, -0.132]	-0.252 [-0.353, -0.152]	-0.021
Late grain-fill	-0.032 [-0.158, +0.094]	-0.043 [-0.170, +0.084]	-0.011

Notes: Entries are percent log-yield changes per additional KDD, with 90% confidence intervals in brackets. The with-GDD specification is the main model and estimates KDD effects conditional on moderate heat accumulation. The no-GDD specification omits GDD while keeping KDD, water-deficit controls, fixed effects, and clustering unchanged. Difference is no-GDD minus with-GDD. All models include municipality fixed effects, agricultural-year fixed effects, and region-specific linear trends, with standard errors clustered by state-year. Source: authors' estimates using the data described in Section 4.

Table A4: Sensitivity of regional KDD estimates to omitting GDD

Region	With GDD	No GDD	Difference
MATOPIBA	-0.096 [-0.214, +0.022]	-0.101 [-0.185, -0.016]	-0.005
MT	+0.011 [-0.041, +0.064]	-0.072 [-0.107, -0.037]	-0.083
GO	-0.197 [-0.273, -0.122]	-0.212 [-0.272, -0.152]	-0.014
MG	-0.131 [-0.222, -0.041]	-0.095 [-0.157, -0.033]	+0.036
MS	-0.110 [-0.205, -0.015]	-0.013 [-0.061, +0.036]	+0.097
SP	-0.026 [-0.099, +0.047]	-0.025 [-0.080, +0.029]	+0.001
PR	-0.145 [-0.223, -0.066]	-0.073 [-0.141, -0.006]	+0.071

Notes: Entries are region-specific phenological window-weighted estimates of the percent log-yield change per additional KDD, with 90% confidence intervals in brackets. Window weights are each region's mean KDD exposure shares across the four yield-cycle windows. The with-GDD specification is the main regional model. The no-GDD specification omits all region-window GDD terms while retaining KDD, water-deficit controls, municipality and agricultural-year fixed effects, region-specific linear trends, and state-year clustering. Difference is no-GDD minus with-GDD. MATOPIBA follows the official 337-municipality definition. Both models use 17,565 observations. Source: authors' estimates using the data described in Section 4.

Data Availability Statement

The public source data used in this study are available from the Brazilian Institute of Geography and Statistics (IBGE PAM), Companhia Nacional de Abastecimento (CONAB), and the gridded weather and auxiliary sources cited in the text (IBGE, 2024; CONAB, 2025; Xavier et al., 2022; Xavier, 2024). Processed analysis data and scripts will be archived in a public replication repository before publication; the repository citation and link will be added once available.

Conflict of Interest

The authors declare no conflicts of interest.

Research Ethics

This study uses publicly available aggregate municipality-level agricultural and weather data and does not involve human participants, animals, or personally identifiable information. Institutional review board approval and informed consent are therefore not applicable.

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